Gateways, BLE Clients and Asynchronous Notifications

Communication Technologies for the Internet of Things, Lesson 06

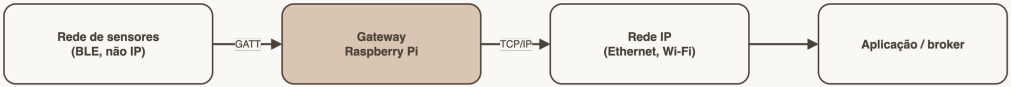
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Agenda

- The role of the gateway in the IoT reference architecture
- Remote administration of a node without peripherals
- The Raspberry Pi in the BLE client role
- Periodic polling and asynchronous notifications
- Forwarding readings to the IP network

The Gateway between Two Networks



O gateway é o único ponto onde os dois mundos de endereçamento se encontram

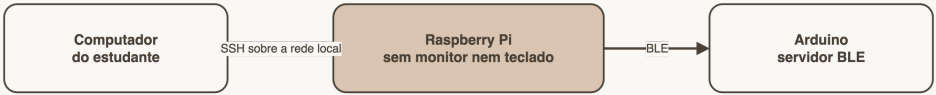
Why a Gateway Is Needed (1/2)

- The short-range technologies studied so far do not use IP addressing: a BLE peripheral is not reachable from the Internet.
- The gateway is the node that participates in both networks simultaneously, translating between the local representation of data and the protocols of the IP network.
- In the reference model presented in Lesson 01, it corresponds to the connectivity level, between the devices and the data processing systems.

Why a Gateway Is Needed (2/2)

- Besides protocol translation, the gateway concentrates functions that constrained nodes cannot take on: aggregation, filtering, temporary storage and security management.
- It is also the point where state can be kept, since it has permanent power and storage.

Remote Administration



A administração remota evita periféricos dedicados para cada Raspberry

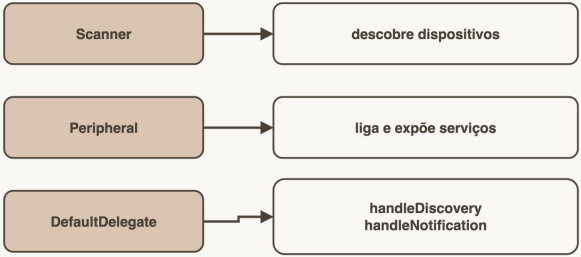
Remote Administration of the Node (1/2)

- In a field-deployed system, the gateway has neither monitor nor keyboard, so all configuration is done remotely.
- SSH access is the usual mechanism: it establishes an encrypted console session over the local network or over the Internet.
- On the Raspberry Pi, the SSH service is enabled by placing an empty file named `ssh` on the boot partition of the card, before first boot.

Remote Administration of the Node (2/2)

- The address assigned to the node on the local network is identified by querying the DHCP server or by scanning the network.
- This approach, termed *headless*, matches the real operating conditions of an IoT deployment.

The bluepy Class Model



O delegado é o ponto onde a aplicação reage a eventos que não controla

The Raspberry Pi as BLE Client (1/2)

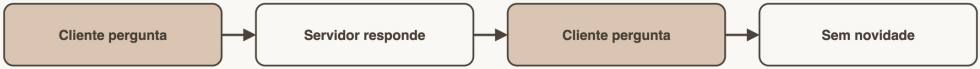
- The Raspberry Pi runs the BlueZ stack, on which the `bluepy` library used in Lesson 05 rests.
- In the central role, it discovers peripherals, establishes connections and queries the services and characteristics exposed.
- The `Scanner`, `Peripheral` and `DefaultDelegate` classes correspond, respectively, to the operations of discovery, connection and reaction to events.

The Raspberry Pi as BLE Client (2/2)

- The delegation pattern is the central point: the application does not control when events occur, merely declaring what to do when they do.
- It is this same pattern that reappears later in subscribing to topics on a message broker.

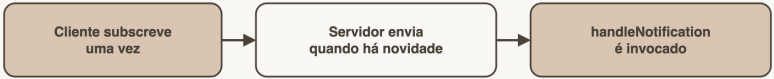
Periodic Polling and Notification

Sondagem periódica



Tráfego mesmo sem dados novos

Notificação assíncrona



Tráfego apenas quando o valor muda

Polling and Notification: A Comparison (1/2)

- In periodic polling, the client queries the characteristic at fixed intervals. Radio traffic is constant and independent of whether new data exists.
- With notification, the client subscribes once and the server transmits only when the value changes.
- The difference matters in battery-powered devices: every unnecessary transmission has a direct energy cost.

Polling and Notification: A Comparison (2/2)

- A notification requires no confirmation from the client, unlike an indication, which is more reliable but generates additional traffic.
- The choice between the two depends on the application's tolerance to the occasional loss of a reading.

Forwarding to the IP Network (1/2)

- Once the notification is received, the gateway must make it available to the systems that consume it, which requires an IP network protocol.
- In this lesson, forwarding is done over a local TCP connection, which keeps the exercise simple and independent of external infrastructure.
- In Lesson 12 this connection is replaced by a message broker, which allows producers and consumers to be decoupled.

Forwarding to the IP Network (2/2)

- The structure of the application nevertheless remains unchanged: the gateway continues to receive notifications and forward them; only the outgoing protocol changes.
- This is why the gateway architecture is addressed before session protocols and message middleware.

Lesson Summary

- The gateway is the only point of contact between the sensor network, which does not use IP, and the data network.
- Remote administration over SSH is the normal way of operating a node without peripherals, and reproduces real operating conditions.
- Notifications replace periodic polling, reducing traffic and consumption, and rest on the same delegation pattern that will be reused in the message middleware.

Sources

Gonçalves, P. *guia 1 — Remote admin*, *guia 3 — BLE Communication* and *aula9 — pythonNotif*, TCIoT, previous editions. Basis for remote administration, the BLE client role and the notifications exercise.

Cisco. *The Internet of Things Reference Model* (2014). Positioning of the gateway at the connectivity level. Available in `Aula 01/Leituras/`.

bluepy library documentation: `ianharvey.github.io/bluepy-doc/`

Raspberry Pi Foundation. Documentation on *headless* configuration and SSH access.

Thank you.

Questions and comments